



Saint Ignatius' College, Riverview

Mathematics Assessment Task

2023

Year 12
Mathematics Extension 1
Task 4
Trial HSC
Date: Friday 18 August 2023

General Instructions:	Exam Sections:																										
• Reading Time: 10 minutes • Time Allowed: 2 hours • Write using black pen • NESA approved calculators may be used • Attempt all questions in the booklets provided. • Use a new Answer Booklet for each of questions 11, 12, 13 and 14. • Write your name and your teacher's code in the positions indicated on the Answer booklets • Marks may not be awarded for missing or carelessly arranged working. Teachers : <table> <tr> <td>• Mr R Maxwell</td><td>REM</td></tr> <tr> <td>• Mr D Reidy</td><td>DPR</td></tr> <tr> <td>• Mr N Mushan</td><td>NHM</td></tr> <tr> <td>• Mr P Collins</td><td>PPC</td></tr> </table>	• Mr R Maxwell	REM	• Mr D Reidy	DPR	• Mr N Mushan	NHM	• Mr P Collins	PPC	<table> <tr> <td>Section A</td><td>10 marks</td></tr> <tr> <td>Multiple Choice</td><td></td></tr> <tr> <td>Section B</td><td></td></tr> <tr> <td>Short Answer</td><td></td></tr> <tr> <td>Question 11</td><td>15 marks</td></tr> <tr> <td>Question 12</td><td>15 marks</td></tr> <tr> <td>Question 13</td><td>15 marks</td></tr> <tr> <td>Question 14</td><td>15 marks</td></tr> <tr> <td>Total</td><td>70 Marks</td></tr> </table>	Section A	10 marks	Multiple Choice		Section B		Short Answer		Question 11	15 marks	Question 12	15 marks	Question 13	15 marks	Question 14	15 marks	Total	70 Marks
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SECTION A: Multiple Choice

Answer Questions 1 to 10 on the Multiple Choice Answer Grid

1. What is the remainder when $P(x) = -2x^3 + 3x^2 - 6x + 4$ is divided by $x + 2$.

- A. 44
- B. -12
- C. 12
- D. 20

2. What is the derivative of $\tan^{-1}(4x-1)$?

- A. $\frac{1}{16x^2 - 8x + 2}$
- B. $\frac{16x - 4}{8x^2 - 4x + 1}$
- C. $\frac{4}{8x^2 - 4x + 1}$
- D. $\frac{2}{8x^2 - 4x + 1}$

3. The domain for $f(x) = 3\cos^{-1} 4x$ is:

- A. $\left[\frac{-3}{4}, \frac{3}{4}\right]$
- B. $\left[\frac{-1}{3}, \frac{1}{3}\right]$
- C. $\left[\frac{-1}{4}, \frac{1}{4}\right]$
- D. $[-1, 1]$

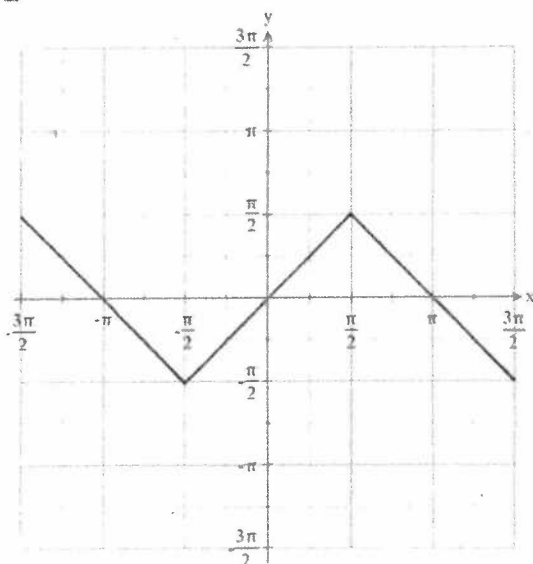
4.	What is the value of $\sin 2x$ given that $\sin x = \frac{\sqrt{3}}{2}$ and x is obtuse? A. $-\frac{\sqrt{3}}{4}$ B. $-\frac{\sqrt{3}}{2}$ C. $\frac{\sqrt{3}}{4}$ D. $\frac{\sqrt{3}}{2}$
5.	A curve C has parametric equations $x = \cos^2 t$ and $y = 2 \sin^2 t$ for $-\infty < t < \infty$. What is the Cartesian equation and domain of C? A. $y = 1 - x$ for $[0, 1]$ B. $y = 2 - 2x$ for $(-\infty, \infty)$ C. $y = 2 - 2x$ for $[0, 1]$ D. $y = 1 - x$ for $(-\infty, \infty)$
6.	Which of the following integrals is equivalent to $\int \cos^2 4x \, dx$? A. $\frac{1}{2} \int (x + \frac{1}{8} \sin 8x) \, dx$ B. $\frac{1}{2} \int (x - \frac{1}{8} \sin 8x) \, dx$ C. $\frac{1}{2} \int (1 + \cos 8x) \, dx$ D. $\frac{1}{2} \int (1 - \cos 8x) \, dx$

7. For the two vectors $\vec{AB}$ and $\vec{AC}$ it is known that $\vec{AB} \cdot \vec{AC} > 0$. Which of the following statements must be true?

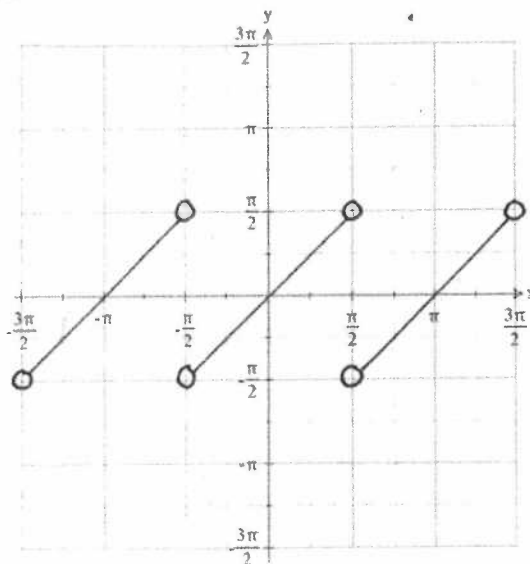
- A. Either $\vec{AB}$ is positive and $\vec{AC}$ is positive, or, $\vec{AB}$ is negative and $\vec{AC}$ is negative.
- B. The angle between $\vec{AB}$ and $\vec{AC}$ is acute.
- C. The product $|\vec{AB}| |\vec{AC}|$ is negative.
- D. The points A , B and C are collinear.

8. Which graph represents the function $y = \tan^{-1}(\tan x)$?

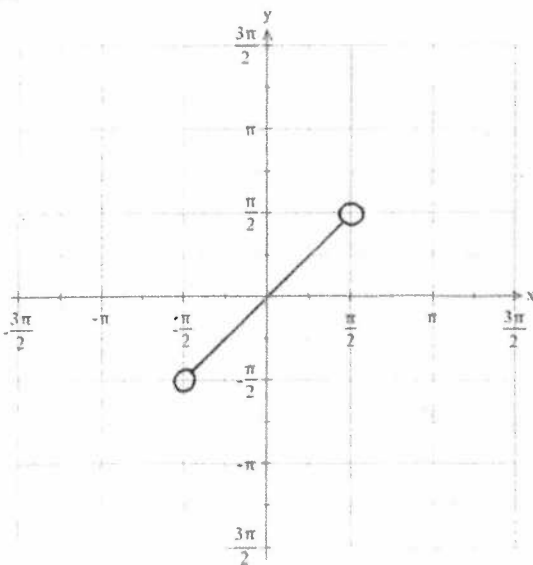
A.



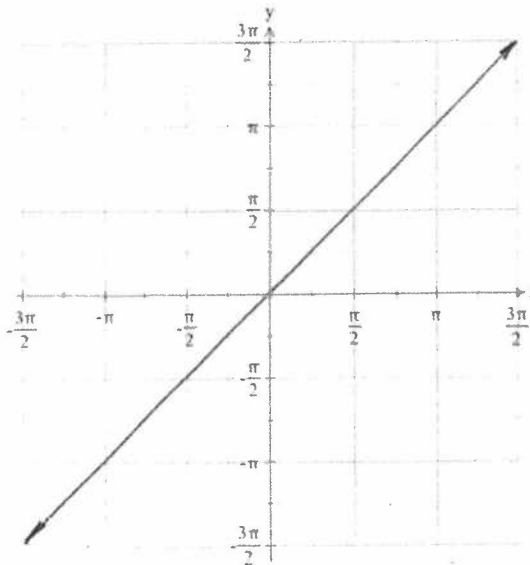
B.



C.



D.

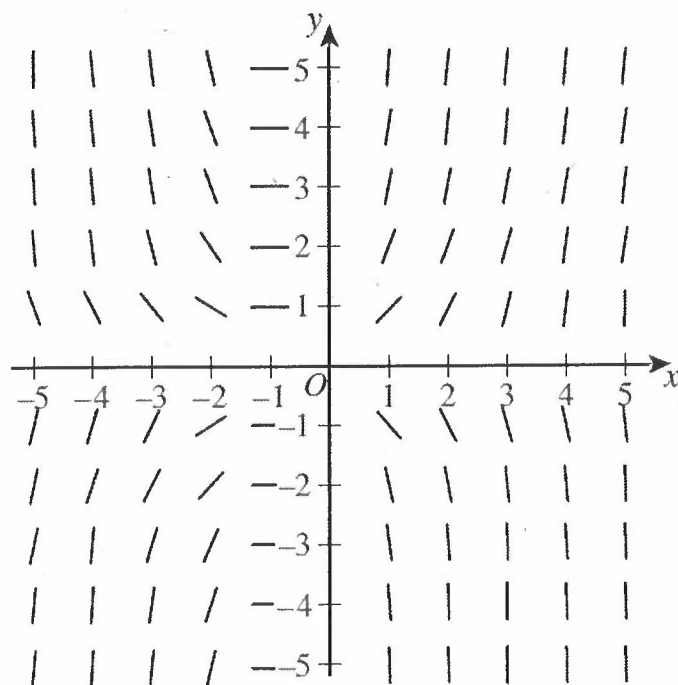


9. Let $P(x) = x^2 + bx + c$ where b and c are constants. The zeros of $P(x)$ are α and $\alpha + 1$.

What are the correct expressions for b and c in terms of α ?

- A. $b = -(2\alpha + 1)$ and $c = \alpha^2 + \alpha$
- B. $b = 2\alpha + 1$ and $c = \alpha^2 + \alpha$
- C. $b = \alpha^2 + \alpha$ and $c = -(2\alpha + 1)$
- D. $b = \alpha^2 + \alpha$ and $c = 2\alpha + 1$

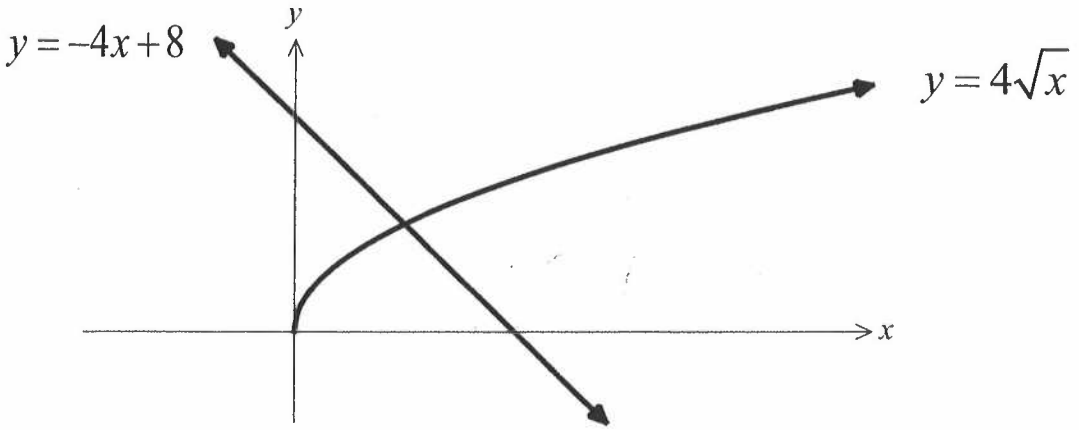
10. The direction (slope) field for a first order differential equation is shown.



Which of the following could be the differential equation represented?

- A. $\frac{dy}{dx} = (x+1)^3$
- B. $\frac{dy}{dx} = x(y+1)$
- C. $\frac{dy}{dx} = (x-1)y$
- D. $\frac{dy}{dx} = (x+1)y$

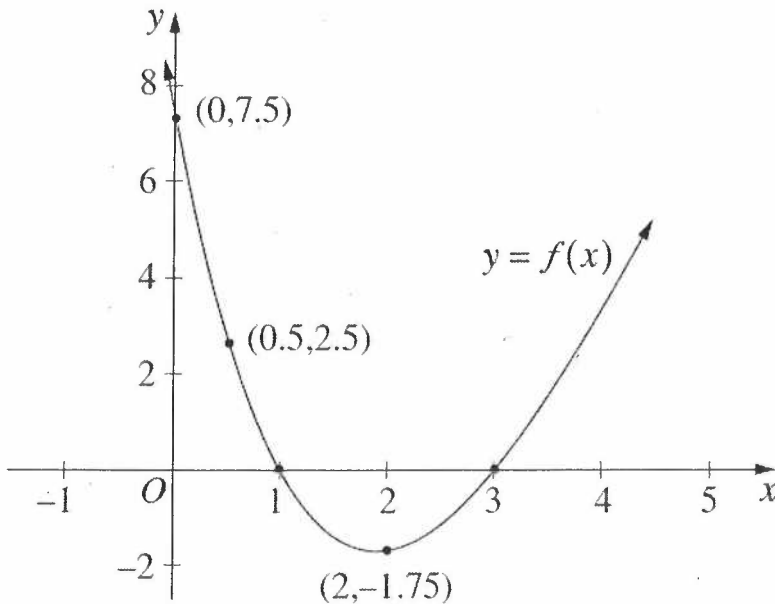
		Marks
a.	Find the anti-derivative of $\frac{1}{\sqrt{4-9x^2}}$	2
b.	i. Find the exact value of $\sin \frac{23\pi}{12}$. (Express answer as a single fraction with a rational denominator)	2
	ii. Find the exact value of $\sin^{-1}(\sin \frac{23\pi}{12})$	1
c.	Consider the vectors $\underline{u} = \underline{i} - 2\underline{j}$ and $\underline{v} = a\underline{i} + 2\underline{j}$ i. Find the value of a if $\underline{u}$ and $\underline{v}$ are perpendicular.	1
	ii. Find the values of a if the angle between $\underline{u}$ and $\underline{v}$ is 60° . Give answers correct to 1 decimal place	3
d.	Given $\underline{u} = \begin{pmatrix} 4 \\ 2 \end{pmatrix}$ and $\underline{v} = \begin{pmatrix} 6 \\ -8 \end{pmatrix}$ i. Find $\text{proj}_{\underline{u}} \underline{v}$	2
	ii. Find the direction of $\text{proj}_{\underline{u}} \underline{v}$ Give answer to nearest degree.	1
e.	Consider the cubic equation $6x^3 - 9x^2 + 4x - 3 = 0$ with roots α , β and γ . Find the value of: i. $\alpha + \beta + \gamma$ ii. $\alpha\beta + \alpha\gamma + \beta\gamma$ iii. $\alpha^2 + \beta^2 + \gamma^2$	1 1 1

		Marks
a.	The area enclosed by the curve $y = 4\sqrt{x}$, the line $y = -4x + 8$ and the x axis is rotated about the x axis.  i. Find the coordinates of the point where $y = 4\sqrt{x}$ and $y = -4x + 8$ intersect. ii. Find the exact volume of the solid of revolution.	1 3
b.	Solve: $\frac{3x}{x+3} \leq -2$	3
c.	Use the substitution $u = 1 + 2 \tan x$ to evaluate $\int_0^{\frac{\pi}{4}} \frac{1}{(1 + 2 \tan x)^2 \cos^2 x} dx$	3
d.	i. Express $\cos x - \sqrt{3} \sin x$ in the form $R \cos(x + \alpha)$ ii. Hence, state the greatest value of the function: $f(x) = \cos x - \sqrt{3} \sin x + 10$	1 1
e.	Consider the function $y = 2x(x+6)^6$. Find the derivative of the inverse function. Give answer in terms of y.	3

QUESTION 13

Use a new writing booklet

(15 marks)

		Marks
a.	The diagram below is a sketch of the graph $y = f(x)$.  i. Sketch the graph of $y = f(x)$ showing all important features, together with the location of the points corresponding to the labelled points on the original sketch. ii. Sketch the graph of $y = \frac{1}{f(x)}$ showing all important features, together with the location of the points corresponding to the labelled points on the original sketch.	2 2
b.	Solve the differential equation $f'(x) = 4x^3 y^2$ given $f(2) = 1$	3
c.	Use mathematical induction to prove $7^{2n} + 7^n + 4$ is divisible by 6 for all integers $n \geq 0$.	3
d.	A metal is heated to 200°C and then left to cool, which it does according to the formula $\frac{dT}{dt} = -0.05(T - 23)$ where T is the temperature in $^\circ\text{C}$ and t is time in minutes. i. Solve the differential equation. (Make T the subject) ii. Find the temperature after 20 minutes. (Answer to 2 decimal places) iii. Find the time when the temperature reaches 40°C. (Answer to 2 decimal places)	3 1 1

		Marks
a.	A particle is projected from a point O on level ground with a speed of 25 ms^{-1} at an angle θ to the horizontal. At time T seconds, the particle passes through the point $B(15, 2)$. The displacement vector $R(t)$ for the motion of the particle is: $R(t) = (Vt \cos \theta) \mathbf{i} + \left(-\frac{1}{2}gt^2 + Vt \sin \theta\right) \mathbf{j} \quad (\text{Do Not Prove this!})$ Where t is the time in seconds after projection, $\mathbf{i}$ and $\mathbf{j}$ are unit vectors of magnitude 1 metre and g is the acceleration due to gravity $= 10 \text{ ms}^{-2}$. i. By considering the horizontal component of the particle's motion, show that $T = \frac{3}{5} \sec \theta$. ii. By considering the vertical component of the particle's motion and, using the result from part i, show that $9 \tan^2 \theta - 75 \tan \theta + 19 = 0$. iii. Find the particle's least possible flight time from O to B. Give your answer correct to 2 decimal places.	1 2 2
b.	Two unbiased dice are rolled. i. Find the probability that the difference of the numbers on the dice is 1. (answer to 2 decimal places) ii. The two dice are rolled 60 times. For the binomial distribution of rolling a difference of 1, find the mean and the variance. (answer to 2 decimal places)	1 2
c.	i. If $t = \tan \frac{A}{2}$, express $3 \sin A - 4 \cos A - 4$ in terms of t ii. Hence solve $3 \sin A - 4 \cos A = 4$ for $0 \leq A \leq 2\pi$ (answer to 2 decimal places)	2 2
d.	Given $\sqrt{\frac{1 - \sin 2x}{1 + \sin 2x}} = \frac{1 - \tan x}{1 + \tan x}$ Show that $\tan \frac{\pi}{8} = \sqrt{2} - 1$	3

END OF TASK



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Suggested Solutions

Exam Sections:		Marker
Section A	10 marks	
Multiple Choice		JPN
Section B: Short Answer		
Question 11	15 marks	PPC
Question 12	15 marks	NHM
Question 13	15 marks	DPR
Question 14	15 marks	REM
Total	70 marks	

Multiple Choice

1. A
2. D
3. C
4. B
5. C
6. C
7. B
8. B
9. A
10. D

Multiple Choice Solutions

1. $P(-2) = -2(-2)^3 + 3(-2)^2 - 6(-2) + 4$
 $= 44$

ANSWER (A)

2. from formula sheet

$$\tan^{-1}(4x-1) = \frac{4}{(1-4x)^2 + 1}$$
$$= \frac{4}{16x^2 - 8x + 1 + 1}$$

$$= \frac{4}{16x^2 - 8x + 2}$$

$$= \frac{2}{8x^2 - 4x + 1}$$

ANSWER (D)

3. $f(x) = 3 \cos^{-1} 4x$

Domain is affected by the 4

$\therefore$ Domain $[-\frac{1}{4}, \frac{1}{4}]$ ANSWER (C)

4. $\sin x = \frac{\sqrt{3}}{2}$ x is obtuse $\therefore \cos x = -\frac{1}{2}$

$$\sin 2x = 2 \sin x \cos x$$

$$= 2 \times \frac{\sqrt{3}}{2} \times -\frac{1}{2}$$

$$= -\frac{\sqrt{3}}{2}$$

ANSWER (B)

5.

$$2x + y = 2 \cos^2 t + 2 \sin^2 t$$
$$= 2(\cos^2 t + \sin^2 t)$$

$$y = 2 - 2x$$

$$\text{Since } x = \cos^2 t \quad 0 \leq x \leq 1$$

$$\therefore y = 2 - 2x \quad [0, 1] \quad \text{ANSWER (C)}$$

6. from reference sheet

$$\cos^2 4x = \frac{1}{2}(1 + \cos 8x)$$

$$\begin{aligned}\therefore \int \cos^2 4x &= \int \frac{1}{2}(1 + \cos 8x) dx \\ &= \frac{1}{2} \int (1 + \cos 8x) dx \quad \text{ANSWER (C)}\end{aligned}$$

7. $\underline{u} \cdot \underline{v} = |\underline{u}| |\underline{v}| \cos \theta$

since $\vec{AB} \cdot \vec{AC} > 0$ and $|\vec{AB}| \times |\vec{AC}| > 0$

$$\therefore \cos \theta > 0$$

$\therefore \theta$ must be acute. ANSWER (B)

8. $\tan^{-1}(\tan x)$ is undefined when $x = \pm \frac{\pi}{2}, \pm \frac{3\pi}{2}, \text{etc}$
but defined for all other values ANSWER (B)

9. $\alpha + \beta = -\frac{b}{a}$

$$\alpha\beta = \frac{c}{a}$$

$$\alpha + \alpha + 1 = -b$$

$$\alpha(\alpha + 1) = c$$

$$2\alpha + 1 = -b$$

$$c = \alpha^2 + \alpha$$

$$\therefore b = -(2\alpha + 1)$$

ANSWER (A)

10. ANSWER (D)

Test numerous points and eliminate

eg. $x = -5$ $y = -1$ gradient positive $\therefore$ not A

$x = -3$ $y = -1$ gradient zero $\therefore$ not B

$x = 1$ $y = 1$ gradient zero $\therefore$ not C

Answer (D)

QUESTION 11

$$\begin{aligned} \textcircled{a} \int \frac{1}{\sqrt{4-9x^2}} dx &= \int \frac{1}{\sqrt{9(\frac{4}{9}-x^2)}} dx \\ &= \int \frac{1}{3\sqrt{\frac{4}{9}-x^2}} dx \\ &= \frac{1}{3} \int \frac{1}{\sqrt{(\frac{2}{3})^2-x^2}} dx \\ &= \boxed{\frac{1}{3} \sin^{-1} \frac{3x}{2} + C} \end{aligned}$$

FROM REF. SHEET.

$$\int \frac{f'(x)}{\sqrt{a^2-(f(x))^2}} dx = \sin^{-1} \frac{f(x)}{a} + C$$

$$\therefore \frac{1}{3} \int \frac{3 \times 1}{\sqrt{2^2-(3x)^2}} = \frac{1}{3} \sin^{-1} \frac{3x}{2} + C$$

Marker's Comments

Many missed the $\frac{1}{3}$ out front or put 3 out front!

$$\begin{aligned} \textcircled{b} \textcircled{i} \sin \frac{23\pi}{12} &= \sin \left(\frac{5\pi}{3} + \frac{\pi}{4} \right) \\ &= \sin \frac{5\pi}{3} \cos \frac{\pi}{4} + \cos \frac{5\pi}{3} \sin \frac{\pi}{4} \\ &= -\frac{\sqrt{3}}{2} \times \frac{1}{\sqrt{2}} + \frac{1}{2} \times \frac{1}{\sqrt{2}} \\ &= \frac{1-\sqrt{3}}{2\sqrt{2}} \\ &= \boxed{\frac{\sqrt{2}-\sqrt{6}}{4}} \end{aligned}$$

OR correct similar

$$\begin{aligned} \textcircled{b} \textcircled{ii} \sin^{-1} \left(\sin \frac{23\pi}{12} \right) &= \sin^{-1} \left(\sin -\frac{\pi}{12} \right) \\ &= \boxed{-\frac{\pi}{12}} \end{aligned}$$

Range of $y = \sin^{-1}$
 $-\frac{\pi}{2} \leq y < \frac{\pi}{2}$

$\therefore$ Not defined for $\frac{23\pi}{12}$ but is for $-\frac{\pi}{12}$.

$$\begin{aligned} \textcircled{c} \textcircled{i} \underline{u} \cdot \underline{v} &= x_1x_2 + y_1y_2 \\ \underline{u} \cdot \underline{v} &= 1 \times a - 2 \times 2 \\ &= a - 4 \end{aligned}$$

perpendicular $\therefore a - 4 = 0$

$$\therefore \boxed{a = 4}$$

$$\begin{aligned} \textcircled{ii} |\underline{u}| &= \sqrt{1+(-2)^2} = \sqrt{5} \\ |\underline{v}| &= \sqrt{a^2+2^2} = \sqrt{a^2+4} \end{aligned}$$

$$\underline{u} \cdot \underline{v} = |\underline{u}| |\underline{v}| \cos \theta$$

$$a - 4 = \sqrt{5} \times \sqrt{a^2+4} \times \cos 60^\circ$$

$$2a - 8 = \sqrt{5a^2+20}$$

$$4a^2 - 32a + 64 = 5a^2 + 20$$

$$0 = a^2 + 32a - 44$$

$$a = \frac{-32 \pm \sqrt{32^2 - 4(1)(-44)}}{2 \times 1}$$

$$\boxed{a = 1.3 \text{ or } -33.3}$$

The New Fx 8200 will do it for you

(i) Very well done

Many problems with algebraic manipulations.

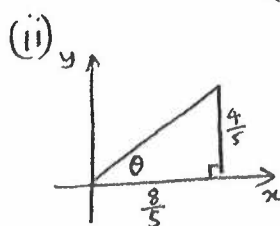
QUESTION 11

Marker's Comments

$$\begin{aligned} \textcircled{d} \text{ (i)} \quad \underline{u} \cdot \underline{v} &= 4 \times 6 + 2 \times -8 \\ &= 24 - 16 \\ &= 8 \end{aligned}$$

$$\begin{aligned} |\underline{u}| &= \sqrt{4^2 + 2^2} \\ &= \sqrt{20} \end{aligned}$$

$$\begin{aligned} \text{proj}_{\underline{u}} \underline{v} &= \frac{\underline{u} \cdot \underline{v}}{|\underline{u}|^2} \underline{u} \\ &= \frac{8}{20} \times \begin{pmatrix} 4 \\ 2 \end{pmatrix} \\ &= \begin{pmatrix} \frac{8}{5} \\ \frac{4}{5} \end{pmatrix} \end{aligned}$$



$$\tan \theta = \frac{4}{5} \div \frac{8}{5}$$

$$\tan \theta = \frac{1}{2}$$

$$\theta = 26.565^\circ$$

$$\text{Direction} = \theta = \boxed{27^\circ}$$

also accept $\frac{2}{5} \begin{bmatrix} 4 \\ 2 \end{bmatrix}$

This formula is not on the Reference sheet. You need to learn it, and apply it, accurately.

Many who were in-correct in part(i) still managed part(ii) as the projection and $\underline{u}$ have the same angle!

$$\textcircled{e} \quad 6x^3 - 9x^2 + 4x - 3 = 0$$

$$\text{(i)} \quad \alpha + \beta + \gamma = -\frac{b}{a} = \frac{9}{6} = \boxed{\frac{3}{2}}$$

$$\text{(ii)} \quad \alpha\beta + \alpha\gamma + \beta\gamma = \frac{c}{a} = \frac{4}{6} = \boxed{\frac{2}{3}}$$

$$\begin{aligned} \text{(iii)} \quad (\alpha + \beta + \gamma)^2 &= \alpha^2 + \alpha\beta + \alpha\gamma + \alpha\beta + \beta^2 + \beta\gamma + \alpha\gamma + \beta\gamma + \gamma^2 \\ &= \alpha^2 + \beta^2 + \gamma^2 + 2(\alpha\beta + \alpha\gamma + \beta\gamma) \end{aligned}$$

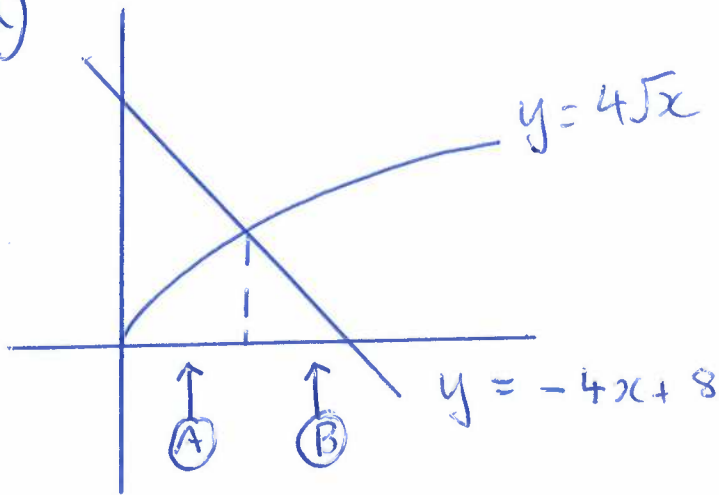
$$\begin{aligned} \therefore \alpha^2 + \beta^2 + \gamma^2 &= (\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \alpha\gamma + \beta\gamma) \\ &= \left(\frac{3}{2}\right)^2 - 2\left(\frac{2}{3}\right) \\ &= \frac{9}{4} - \frac{4}{3} \\ &= \boxed{\frac{11}{12}} \end{aligned}$$

✓ Page 1 of Reference sheet!
✓ (i)(ii) Generally well done

(iii) No so well done

QUESTION 12

a)



$$\begin{aligned}
 \text{i) } y &= 4\sqrt{x} & 4\sqrt{x} &= -4x + 8 \\
 y &= -4x + 8 & 16x &= 16x^2 - 64x + 64 \\
 & & 0 &= 16x^2 - 80x + 64 \\
 & & &= x^2 - 5x + 4 \\
 & & &= (x-1)(x-4)
 \end{aligned}$$

$$\begin{aligned}
 \therefore x &= 1 & x &= 4 \\
 y &= 4 & y &= -8
 \end{aligned}$$

$\therefore$ Point of Intersection

$$(1, 4) \leftarrow \text{1mk}$$

$$\text{ii) } V = \pi \int_a^b y^2 dx$$

$$\begin{aligned}
 \text{Volume (A) } V &= \pi \int_0^1 16x dx \\
 &= \pi [8x^2]_0^1 \\
 &= 8\pi \text{ units}^3 \leftarrow \text{1mk}
 \end{aligned}$$

Fairly well done.

Some careless errors.

Volume (B)

$$V = \pi \int_1^2 (-4x + 8)^2 dx$$

$$= \pi \int_1^2 (16x^2 - 64x + 64) dx$$

$$= \pi \left[\frac{16x^3}{3} - 32x^2 + 64x \right]_1^2$$

$$= \pi \left[\left(\frac{128}{3} \right) - \left(\frac{112}{3} \right) \right]$$

$$= \frac{16\pi}{3} \text{ Units}^3 \quad \leftarrow \text{1mk}$$

$$\text{Volume Total} = 8\pi + \frac{16\pi}{3}$$

$$= \frac{40\pi}{3} \text{ u}^3 \quad \leftarrow \text{1mk}$$

(4)

b) $\frac{3x}{x+3} \leq -2$

$$3x(x+3) \leq -2(x+3)^2$$

$$2(x+3)^2 + 3x(x+3) \leq 0$$

$$(x+3)(2(x+3) + 3x) \leq 0$$

$$(x+3)(5x+6) \leq 0$$



$$-3 < x \leq -\frac{6}{5}$$

Many need more time revising the volumes topic.

Too many had forgotten method.

$$-3 < x \leq -\frac{6}{5}$$

↑
1mk
↑
1mk

↑
1mk

(3)

c)

$$\int_0^{\frac{\pi}{4}} \frac{1}{(1+2\tan x)^2 \cos^2 x} dx$$

$$u = 1 + 2\tan x$$

$$\frac{du}{dx} = 2\sec^2 x$$

$$du = 2\sec^2 x dx$$

$$\frac{1}{2} du = \frac{1}{\cos^2 x} dx$$

$$x=0 \quad u=1$$

$$x=\frac{\pi}{4} \quad u=3$$

poorly done

$$\therefore \int_0^{\frac{\pi}{4}} \frac{1}{(1+2\tan x)^2 \cos^2 x} dx = \frac{1}{2} \int_1^3 \frac{1}{u^2} du$$

$$= \frac{1}{2} \int_1^3 u^{-2} du$$

$$= \frac{1}{2} \left[-\frac{1}{u} \right]_1^3$$

$$= \frac{1}{2} \left[\left(-\frac{1}{3} \right) - (-1) \right]$$

$$= \frac{1}{3}$$

mk for correct integral

mk.

3

$$d) \quad 1) \cos x - \sqrt{3} \sin x \\ = R \cos(x + \alpha)$$



$$R^2 = 1^2 + (\sqrt{3})^2 \quad \tan \alpha = \frac{\sqrt{3}}{1} \\ R = 2 \quad \alpha = \frac{\pi}{3}$$

$$\therefore \cos x - \sqrt{3} \sin x = 2 \cos\left(x + \frac{\pi}{3}\right) \leftarrow \text{mk}$$

$$1) \quad f(x) = \cos x - \sqrt{3} \sin x + 10 \\ = \underbrace{2 \cos\left(x + \frac{\pi}{3}\right)}_{\text{max} = 2} + 10$$

$$\therefore \text{Max} = 12 \quad \leftarrow \text{mk}$$

2

e)

$$y = 2x(x+6)^6$$

$$x = 2y(y+6)^6 \leftarrow \text{Imk}$$

$$\begin{aligned} \frac{dx}{dy} &= 2y[6(y+6)^5] + [y+6]^6 \cdot 2 \leftarrow \text{Imk for} \\ &= (y+6)^5 \cdot (12y + 2(y+6)) \end{aligned}$$

Correct use of product rule

$$= (y+6)^5 \cdot (14y + 12)$$

$$= 2(y+6)^5(7y+6)$$

$$\therefore \frac{dy}{dx} = \frac{1}{2(y+6)^5(7y+6)} \leftarrow \text{Imk}$$

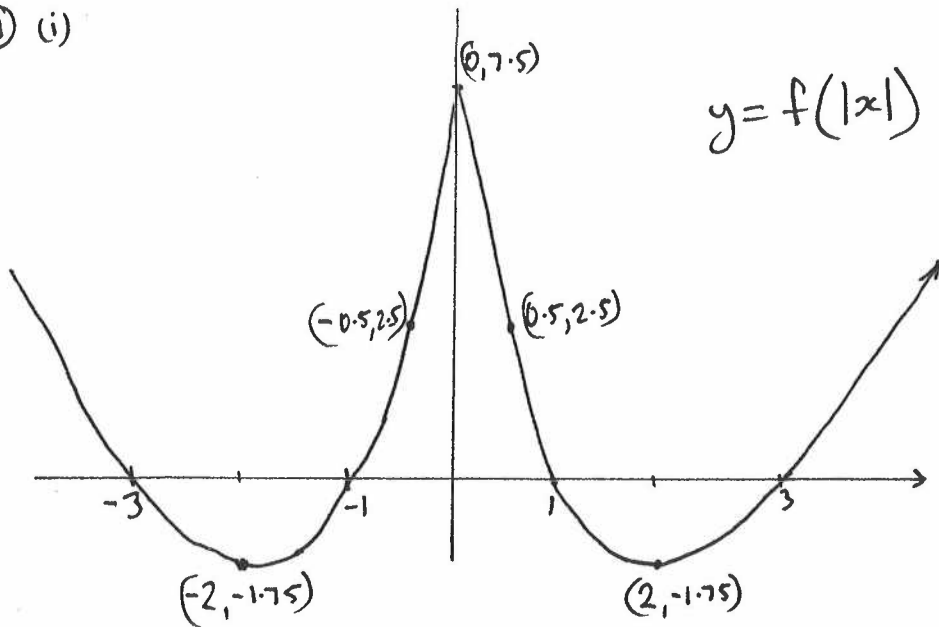
Desirable to go to $\frac{dy}{dx} =$

but no penalty if left at
CORRECT $\frac{dx}{dy}$

QUESTION 13

Marker's Comments

(a) (i)

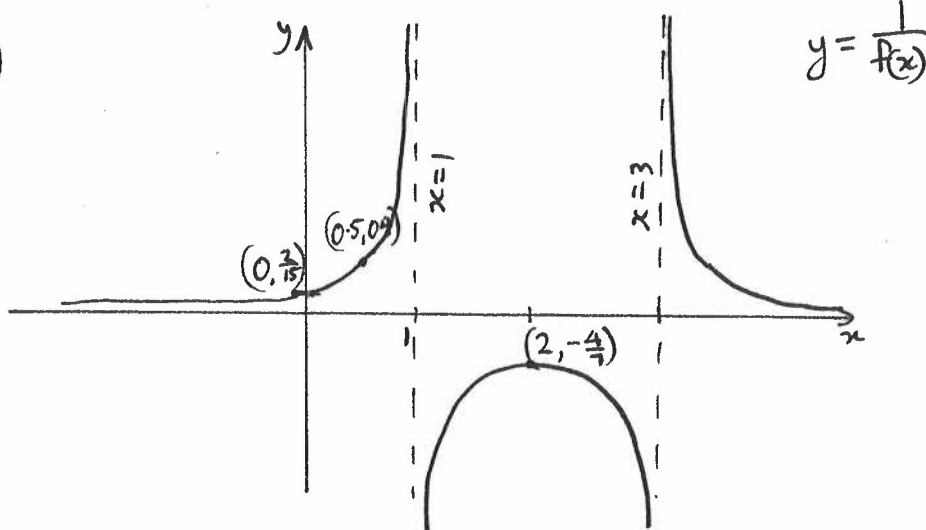


* Well answered.

* 1 mark for correct shape.

* 1 mark for including coordinates of intercepts and turning points.

(ii)



* Well answered

* 1 mark for correct shape

* 1 mark for including coordinates of intercepts and turning points.

(b) $\frac{dy}{dx} = 4x^3 y^2$

$$\frac{1}{y^2} dy = 4x^3 dx$$

$$\int y^{-2} dy = \int 4x^3 dx$$

$$\frac{y^{-1}}{-1} + c_1 = x^4 + c_2$$

$$-\frac{1}{y} = x^4 + k$$

when $x=2, y=1$

$$-\frac{1}{1} = 2^4 + k$$

$$-1 = 16 + k$$

$$k = -17$$

$$\therefore -\frac{1}{y} = x^4 - 17$$

$$y = \frac{1}{17 - x^4}$$

* Well answered.

* some students did not calculate the constant correctly but received carry on marks.

* some students did not set up the differential correctly at the start but received carry on marks afterwards.

QUESTION 13

Marker's Comments

① Prove $7^{2n} + 7^n + 4$ is divisible by 6 for all integers $n \geq 0$

Step 1: Prove true for $n=0$

$$7^{2 \times 0} + 7^0 + 4 = 1 + 1 + 4 = 6 \text{ which is divisible by 6}$$

$$\therefore \text{true for } n=0$$

Step 2: Assume true for $n=k$

$$\text{ie } 7^{2k} + 7^k + 4 = 6M \text{ where } M \text{ is a positive integer}$$

$$\text{ie } 7^{2k} = 6M - 7^k - 4$$

Step 3: Prove true for $n=k+1$

$$\text{ie } 7^{2(k+1)} + 7^{(k+1)} + 4 = 6P \text{ where } P \text{ is a positive integer}$$

$$\text{LHS} = 7^{2k+2} + 7^{k+1} + 4$$

$$= 7^{2k} \times 7^2 + 7^{k+1} + 4$$

$$= 49(6M - 7^k - 4) + 7 \times 7^k + 4 \dots \text{from assumption in step 2}$$

$$= 294M - 49 \times 7^k - 196 + 7 \times 7^k + 4$$

$$= 294M - 42 \times 7^k - 192$$

$$= 6(49M - 7 \times 7^k - 32)$$

$$= 6P \text{ since } M \text{ and } k \text{ are integers.}$$

If true for $n=k$, then true for $n=k+1$

Since it is true for $n=0$, then it is true for $n=1$.

If true for $n=1$, then it is true for $n=2$ and so on
for all positive integers $n \geq 0$

* Well answered.

* Some students

substituted $n=1$

for the first step

and got penalised.

QUESTION 13

Marker's Comments

$$(d) (i) \frac{dT}{dt} = -0.05(T-23)$$

$$\frac{dt}{dT} = \frac{-1}{0.05(T-23)}$$

$$= \frac{-20}{T-23}$$

$$t = -20 \int \frac{1}{T-23} dT$$

$$= -20 \ln|T-23| + c$$

$$\text{when } t=0 \quad T=200$$

$$0 = -20 \ln|200-23| + c$$

$$c = 20 \ln 177$$

$$\therefore t = -20 \ln|T-23| + 20 \ln 177$$

$$t = 20(\ln 177 - \ln|T-23|)$$

$$t = 20 \ln \frac{177}{T-23}$$

$$\frac{t}{20} = \ln \frac{177}{T-23}$$

$$e^{\frac{t}{20}} = e^{\ln \frac{177}{T-23}}$$

$$\frac{177}{T-23} = e^{\frac{t}{20}}$$

$$177 = (T-23)e^{\frac{t}{20}}$$

$$177e^{-\frac{t}{20}} = T-23$$

$$T = 23 + 177e^{-\frac{t}{20}}$$

$$T = 23 + 177e^{-0.05t}$$

$$(ii) \text{ when } t=20$$

$$T = 23 + 177e^{-0.05 \times 20}$$

$$T = 88.11^\circ \text{C}$$

$$(iii) \text{ when } T=40$$

$$40 = 23 + 177e^{-0.05t}$$

$$17 = 177e^{-0.05t}$$

$$\frac{17}{177} = e^{-0.05t}$$

$$\ln \frac{17}{177} = -0.05t$$

$$t = \frac{\ln \frac{17}{177}}{-0.05} = 46.86 \text{ minutes}$$

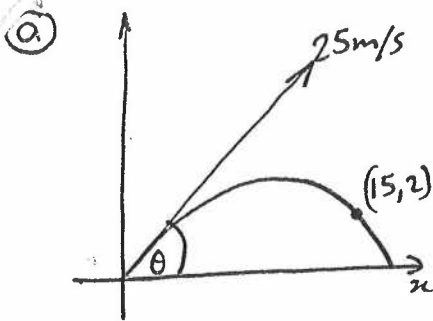
46 minutes 52 seconds

* Well answered

* Some students incorporated the ~~power~~ constant of integration into the exponential power when calculating and lost marks.

* Well answered

* Well answered.



When $t = T$
 $x = 15$

(i) $x = Vt \cos \theta$
 $15 = 25t \cos \theta$
 $T = \frac{15}{25 \cos \theta}$ ✓

$T = \frac{3}{5} \sec \theta$ (as required)

(ii) $y = Vt \sin \theta - \frac{1}{2} \times 10 \times t^2$

When $t = T$
 $y = 2$

$2 = 25 \times \frac{3}{5} \sec \theta \sin \theta - 5 \left(\frac{3}{5} \sec \theta \right)^2$

$2 = 15 \tan \theta - 5 \times \frac{9}{25} \sec^2 \theta$ → ✓ 1 Mark

$2 = 15 \tan \theta - \frac{9}{5} \sec^2 \theta$

$10 = 75 \tan \theta - 9(1 + \tan^2 \theta)$ → ✓ 1 Mark

$10 = 75 \tan \theta - 9 - 9 \tan^2 \theta$

$9 \tan^2 \theta - 75 \tan \theta + 19 = 0$
 (as required)

(iii) $\tan \theta = \frac{75 \pm \sqrt{75^2 - 4(9)(19)}}{2 \times 9}$
 $= \frac{75 \pm \sqrt{4941}}{18}$

$\tan \theta = 0.2615418...$ or $\tan \theta = 8.07179...$ → 1 Mark
 $\theta = 14.6569...^\circ$ or $\theta = 82.9377...^\circ$

$\therefore T = \frac{3}{5 \cos 14.656...^\circ}$
 $= 0.62 \text{ seconds}$
 (2d.p)

$T = \frac{3}{5 \cos 82.9377...^\circ}$
 $= 4.88 \text{ seconds}$

$\therefore$ Least possible flight time from O to B

is 0.62 seconds ✓

Correct Answer.

QUESTION 14

Marker's Comments

(b)

(i) 2nd Die	1st Die		Difference					
	1	2	3	4	5	6		
1	0	1	2	3	4	5		
2	1	0	1	2	3	4		
3	2	1	0	1	2	3		
4	3	2	1	0	1	2		
5	4	3	2	1	0	1		
6	5	4	3	2	1	0		

$$P(\text{difference} = 1) = \frac{10}{36} = \boxed{\frac{5}{18}} \text{ OR } 0.28 \rightarrow 1 \text{ Mark}$$

$$(ii) \text{ mean} = E(X) = 60 \times \frac{5}{18} = \frac{50}{3} = 16\frac{2}{3} = \boxed{16.6} \text{ OR } 60 \times 0.28 \rightarrow \checkmark$$

$$= 16.67 \quad = 16.80 \rightarrow \checkmark$$

$$(iii) \text{ Variance} = \text{Var}(X) = np(1-p)$$

$$= 60 \times \frac{5}{18} \left(1 - \frac{5}{18}\right) \text{ OR } 60 \times 0.28 \times (1 - 0.28) \rightarrow \checkmark$$

$$= \frac{325}{21} \quad = 12.10 \rightarrow \checkmark$$

$$= \boxed{12.04} \text{ (2 d.p.)}$$

QUESTION 14

Marker's Comments

① $3 \sin A - 4 \cos A - 4$

$$= 3 \times \frac{2t}{1+t^2} - 4 \times \frac{(1-t^2)}{1+t^2} - 4$$

$$= \frac{6t - 4 + 4t^2 - 4 - 4t^2}{1+t^2}$$

$$= \frac{6t-8}{1+t^2}$$

② $3 \sin A - 4 \cos A = 4$

$$3 \sin A - 4 \cos A - 4 = 0$$

$$\therefore \frac{6t-8}{1+t^2} = 0$$

$$6t - 8 = 0$$

$$6t = 8$$

$$t = \frac{4}{3}$$

$$\tan \frac{A}{2} = \frac{4}{3}$$

$$\frac{A}{2} = 0.9272$$

$$A = 1.85 \text{ (2d.p.)}$$

$$0 \leq A \leq 2\pi$$

$$\therefore 0 \leq \frac{A}{2} \leq \pi \text{ (only 1st Q)}$$

* NOTE: Need to

Check $A = \pi$ as
a solution ✓

$$\begin{aligned} \text{LHS} &= 3 \sin \pi - 4 \cos \pi \\ &= 0 + 4 = 4 \\ &= \text{RHS} \end{aligned}$$

$\therefore A = \pi$ is a solution

1 Mark Correct answer

③ $\sqrt{\frac{1-\sin 2x}{1+\sin 2x}} = \frac{1-\tan x}{1+\tan x}$

$$\text{let } x = \frac{\pi}{8} \therefore 2x = \frac{\pi}{4} \quad (\text{since } \tan \frac{\pi}{4} = 1)$$

$$\sqrt{\frac{1-\sin \frac{\pi}{4}}{1+\sin \frac{\pi}{4}}} = \frac{1-\tan \frac{\pi}{8}}{1+\tan \frac{\pi}{8}} \rightarrow 1 \text{ Mark}$$

$$\sqrt{\frac{1-\frac{1}{\sqrt{2}}}{1+\frac{1}{\sqrt{2}}}} = \frac{\tan \frac{\pi}{4} - \tan \frac{\pi}{8}}{1 + \tan \frac{\pi}{4} \tan \frac{\pi}{8}} \rightarrow 1 \text{ Mark}$$

$$\sqrt{\frac{\sqrt{2}-1}{\sqrt{2}+1}} = \tan\left(\frac{\pi}{4} - \frac{\pi}{8}\right)$$

$$\sqrt{\frac{\sqrt{2}-1}{\sqrt{2}+1} \times \frac{\sqrt{2}-1}{\sqrt{2}-1}} = \tan \frac{\pi}{8}$$

$$\sqrt{\frac{(\sqrt{2}-1)^2}{2-1}} = \tan \frac{\pi}{8}$$

$$\sqrt{2}-1 = \tan \frac{\pi}{8}$$

as required.

1 Mark for correct simplification of right hand side.

Alternatively!

$$\sqrt{2}-1 = \frac{1-\tan \frac{\pi}{8}}{1+\tan \frac{\pi}{8}}$$

$$\sqrt{2} + \sqrt{2} \tan \frac{\pi}{8} - 1 - \tan \frac{\pi}{8} = 1 - \tan \frac{\pi}{8}$$

$$\sqrt{2} \tan \frac{\pi}{8} = 2 - 2\sqrt{2}$$

$$\tan \frac{\pi}{8} = \frac{2-2\sqrt{2}}{\sqrt{2}}$$

$$= \sqrt{2}-1$$

* Can square both sides & form a quadratic in $\tan \frac{\pi}{8}$